

Adrin N Makombe

AI Automation Engineer & Full-Stack Developer | Blockchain · SaaS · AI Systems · Linux Platform Engineering

- Cape Town, South Africa · Remote Available
- amakodev@gmail.com
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- Contract / Full-time Available
- Dev Career Since 2021 · Digital Business Background: 2017
- amakodev.github.io/portfolio
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PROFESSIONAL SUMMARY

AI Automation Engineer and full-stack developer with a 4-year software development career (2021–present), progressing rapidly from blockchain roots through AI automation, SaaS product delivery, cross-platform mobile, and Linux-level OS engineering. Career arc: *Blockchain Developer at Okamalabs (2021)* → *International Freelance Web3 & AI on Upwork (2023)* → *AI Systems Developer at USCreativz (Sep 2024)* → *Co-Founded Zyntrix Solutions (2025)* → *Linux Kernel Engineering for OkamaOS gaming platform using Buildroot (April 2026)*. Prior to software development, built 4 years of commercial experience across digital marketing, content creation, e-commerce, and affiliate marketing (2017–2021) — providing the product intuition and business context that informs product-first engineering decisions. Operates under the **amakodev** personal brand throughout the development career. Maintains **130+ repositories across 24+ product categories**. Strongest fit for full-stack, AI automation, product engineering, or platform-adjacent roles.

Dev Career: 2021–Present

Blockchain → AI Automation → Linux OS

130+ Repositories · 24+ Categories

Co-Founded Zyntrix Solutions (2025)

React · Next.js · React Native · Flutter

Contract / Full-time Open

TECHNICAL SKILLS

FRONTEND

React

Next.js

TypeScript

JavaScript

Vite

Tailwind CSS

Framer Motion

React Query

Zustand

HTML5

CSS3

BACKEND & APIS

Node.js

Python

REST APIs

API Integrations

API Gateway Design

Backend Services

Database Design

Axios

BLOCKCHAIN & WEB3

Blockchain Development

Smart Contracts

Web3

Decentralised Systems

dApp Development

Cryptographic Protocols

ARCHITECTURE & SYSTEMS

SaaS Architecture

System Architecture

Platform Design

Product Ecosystem Design

Modular Architecture

API Proxy Design

Event-Driven Architecture

Git / GitHub

Vitest

Jest

LINUX & OS SYSTEMS

Linux

Buildroot

Kernel Engineering

Custom Linux Distribution

Embedded Linux

Boot Process Configuration

OS Architecture

Cross-Compilation

Root Filesystem Design

Linux Gaming Platform

Init System Configuration

Kernel Module Selection

MOBILE

React Native

Flutter

Expo

Expo Router

EAS Build

Cross-Platform Dev

Zustand Persist

AI & AUTOMATION

AI Agents

LLM API Integrations

Workflow Automation

Automation Pipelines

Google Apps Script

MQL5 / MetaTrader 5

CLOUD & DATA

Firebase

Firestore

Firebase Auth

Firebase Realtime DB

Expo SecureStore

Google Sheets API

Vercel

PROFESSIONAL EXPERIENCE

Co-Founder & Platform Engineer

2025 – Present

[Zyntrix Solutions — Digital Systems & Technology Services · Remote \(Co-founded & rebranded from SoftDevZA\)](#)

- ▶ Co-founded Zyntrix Solutions, scaling from SoftDevZA into a structured digital systems company delivering web, mobile, AI, and platform-level engineering.
- ▶ Engineered OkamaOS (April 2026): a custom Linux gaming platform built using Buildroot as the core build system — configuring kernel modules, establishing a cross-compilation environment, generating a gaming-optimised root filesystem, and designing the full OS distribution from the ground up.
- ▶ Configured Linux kernel for gaming-specific requirements: graphics subsystem integration, input device support, hardware compatibility, and boot process optimisation for a dedicated gaming environment.
- ▶ Developed ZyntrixOS: an OS-inspired workspace platform delivering a browser-native product shell with unified apps, dashboards, and developer workflows above the Linux platform layer.
- ▶ Maintained a 7-repository ecosystem (okamaos, okamaos-studio, zyntrixos and others) spanning the OS kernel layer, companion workspace tooling, and web platform product.

Stack: Linux · Buildroot · Kernel Engineering · Custom OS Distribution · React · TypeScript · Node.js · Platform Architecture

Software Developer — Web, Mobile & AI

2025

[SoftDevZA — Software Development Agency \(later co-founded & rebranded to Zyntrix Solutions\)](#)

- ▶ Delivered web, mobile, and AI development projects for client platforms — expanding technical range into cross-platform mobile delivery with React Native and Flutter alongside ongoing web/AI work.
- ▶ Built AI-integrated features and workflow automation systems for client products, applying practical AI tooling to real production environments.
- ▶ Contributed to scalable SaaS architecture and platform design work that directly informed the decision to co-found and rebrand to Zyntrix Solutions.

Stack: React · React Native · Flutter · Node.js · Python · REST APIs · AI Systems · JavaScript · TypeScript

Developer — Web & AI Systems

Sep 2024 – 2025

[USCreativz — Digital Marketing & Technology Agency](#)

- ▶ Transitioned from freelance into a structured agency environment focused on web development and AI systems — bridging digital marketing operations with technical product delivery.
- ▶ Built web-based tools and AI-assisted automation systems supporting campaign management, internal workflow tooling, and digital delivery operations for marketing teams.
- ▶ Applied prior digital marketing background to build product-aware tools that mapped directly to real marketing team workflows and operational pain points, improving delivery outcomes.

Stack: JavaScript · React · Node.js · AI Automation · Workflow Systems · APIs · Web Applications

Freelance Developer — Blockchain & Web3 to AI Automation

2023 – Sep 2024

[amakodev](#) · [Upwork](#) — [International Clients](#) · [Remote](#)

- ▶ Launched international freelance career on Upwork immediately post-graduation, securing blockchain and Web3 projects with clients across international markets.
- ▶ Delivered Neptune Chain: an international blockchain client project applying smart contract and Web3 development in a production-facing environment.
- ▶ Progressively pivoted from blockchain/Web3 into web development, AI automation, and workflow systems — tracking market demand and expanding technical delivery range across client engagements.
- ▶ Operated under the amakodev brand as the professional identity for all freelance and product development work — a brand carried continuously from the start of the development career through all subsequent roles.

Stack: [Blockchain](#) · [Smart Contracts](#) · [Web3](#) · [React](#) · [Node.js](#) · [REST APIs](#) · [AI Automation](#) · [Workflow Automation](#)

Blockchain Developer

Mid 2021 – 2023

[Okamalabs](#) — [Blockchain & Technology](#) (during undergraduate degree)

- ▶ First professional development role — joined Okamalabs during undergraduate study at Richfield Graduate Institution, establishing the foundation of a software engineering career with a hands-on focus on blockchain and decentralised systems.
- ▶ Developed blockchain-based applications and smart contract systems, gaining deep technical grounding in decentralised architecture, cryptographic protocols, and Web3 development patterns.
- ▶ Ran development work in parallel under the amakodev brand from the start, combining academic coursework with real-world delivery that shaped the product thinking carried into all subsequent roles.

Stack: [Blockchain](#) · [Smart Contracts](#) · [Web3](#) · [Decentralised Systems](#) · [Programming Fundamentals](#) · [Software Engineering](#)

FLAGSHIP PROJECTS

Portfolio Note: GitHub repository tracking began in 2024. Projects below are recent documented work from an active 4-year development career (2021–present) spanning blockchain, AI automation, SaaS, mobile, and Linux platform engineering.

ViralFlow Ink — AI Content Operations SaaS

SAAS · ACTIVE

AI-assisted workflow platform centralizing content planning, scheduling, and campaign execution for creator and marketing teams.

- ▶ Designed an AI-assisted workflow SaaS centralizing content planning, scheduling, and execution for faster, more repeatable campaign operations.
- ▶ Engineered a React 18 + TypeScript frontend with reusable workflow patterns, structured scheduling views, and product-grade UX optimised for team coordination.
- ▶ Architected integration-heavy backend and API proxy flows connecting operational workflows to Google Sheets and external services, reducing tool-switching and manual handoffs.
- ▶ Owned delivery end-to-end: product design → frontend engineering → systems architecture → deployment → automation feature design.

Stack: React 18 · TypeScript · Vite · Tailwind · Firebase · Node.js · Google Apps Script · Google Sheets API · Vercel **Focus:** AI workflow design · SaaS product architecture · API integrations · end-to-end product ownership

OkamaOS — Custom Linux Gaming Platform & ZyntrixOS Workspace Ecosystem

LINUX · KERNEL ENGINEERING · APR 2026

Custom Linux gaming platform built from the ground up using Buildroot — kernel configuration, gaming-optimised OS distribution design, hardware integration, and a 7-repository platform ecosystem extending into web-delivered workspace tooling.

- ▶ Engineered OkamaOS: a custom Linux gaming platform using **Buildroot** as the core build system — configuring kernel modules, setting up a cross-compilation environment, generating a gaming-optimised root filesystem, and designing the full OS distribution from the ground up.
- ▶ Configured Linux kernel parameters for gaming-specific requirements: hardware compatibility layers, graphics subsystem integration, input device support, and boot process optimisation for a dedicated gaming environment.
- ▶ Developed OkamaOS Studio as a companion workspace and developer tooling layer above the OS core — providing application management, developer workflows, and a product shell interface running on top of the custom Linux base.
- ▶ Extended the same systems thinking into ZyntrixOS: a web-delivered OS-style workspace platform unifying apps, dashboards, tools, and developer workflows into a cohesive browser-native operating environment.
- ▶ Maintained a 7-repository platform ecosystem (amakodev/okamaos, amakodev/okamaos-studio, amakodev/okamaos_projectX, amakodev/zyntrixos, amakodev/trust-os-hub, zyntrixsolutions/ZyntrixOS, zyntrixsolutions/okamaos) spanning kernel builds, workspace layers, and platform tooling.

Stack: Linux · Buildroot · Kernel Engineering · Custom OS Distribution · Root Filesystem Design · Cross-Compilation · React · TypeScript · Platform Architecture **Focus:** Kernel engineering · Linux gaming platform · embedded Linux · OS distribution design · workspace product architecture · platform ecosystem design

MiClass — Personalized AI Tutor App

MOBILE · AI-ENABLED

Production-grade mobile learning application with AI-enabled tutoring flows, secure data handling, and version-controlled rollout.

- ▶ Developed a React Native / Expo application with authentication, secure local storage, Firebase-backed data flows, and structured document handling for learner-facing experiences.
- ▶ Integrated AI endpoint configuration and PDF/document processing capabilities to support personalized tutoring workflows and smarter in-app learning assistance.
- ▶ Implemented version-gating and configuration controls improving rollout safety and product operations discipline in a production mobile environment.

Stack: Expo · React Native · Firebase Auth · Firebase Realtime DB · Expo SecureStore · Axios · Document Picker · EAS Build **Focus:** Mobile product delivery · AI-enabled edtech · secure storage · backend integration

JusAesthetic — Realtime Cross-Platform Social App

MOBILE · REALTIME

Authenticated realtime social experience with Firestore subscriptions, device-native camera/QR integrations, and TypeScript Expo Router architecture.

- ▶ Built a TypeScript Expo Router application with authenticated route gating, profile-aware flows, and realtime user-state updates via Firestore subscriptions.
- ▶ Integrated social/profile data patterns and device capabilities including camera, QR scanning, and authentication-linked user experiences.
- ▶ Structured routing and state logic for consistent UI behaviour across changing auth states, improving continuity in a realtime mobile environment.

Stack: Expo · TypeScript · Expo Router · React Native Firebase · Firestore · Google Sign-In · Camera/QR · Jest **Focus:** Realtime app engineering · cloud-backed state · cross-platform development · mobile UX systems

Grind City Empire — Browser-Based Web Game

ACTIVE BUILD

Complex frontend state management and progression-system design demonstrating advanced product engineering depth beyond standard CRUD applications.

- ▶ Developed a browser-based game with multiple screens, persisted state, upgrade and economy mechanics, and progression logic to support deeper user interaction loops.
- ▶ Used Zustand Persist and modular stores to handle complex gameplay behaviour, decision flows, and reusable front-end architecture beyond standard CRUD applications.
- ▶ Included Vitest testing configuration and modular screen design, reinforcing maintainability and engineering discipline in a richer interactive product.

Stack: React · TypeScript · Vite · Tailwind · Zustand Persist · React Query · Framer Motion · Vitest **Focus:** State-heavy frontend systems · progression logic · modular architecture · testing discipline

ScalpGoat — Algorithmic Trading System Experiment

R&D · EXPERIMENT

Rules-driven trading automation experiment demonstrating analytical thinking, modular signal processing, and event-driven systems design outside traditional stacks.

- ▶ Developed a MetaTrader 5 expert advisor with modular components for indicators, market condition scoring, and execution constraints.
- ▶ Combined multi-timeframe signal logic, support/resistance analysis, and risk-control concepts into a structured automated decision system.
- ▶ Separated analytical logic, conditions, and trade behaviour into modular units for cleaner maintainability — demonstrating comfort building algorithmic systems outside traditional web and mobile stacks.

Stack: MQL5 · MetaTrader 5 · Algorithmic Strategy Logic · Event-Driven Rules · Risk Constraints **Focus:** Analytical systems design · automation logic · rule-based architectures · event-driven programming

EDUCATION

Bachelor of Science in Information Systems

2021 – 2023

Richfield Graduate Institution · South Africa

- ▶ Core study areas: programming, web technology, information systems, databases, systems analysis, and digital business problem-solving.
- ▶ Strengthened technical perspective through coursework in business systems, data management, and practical IT problem-solving.
- ▶ Worked concurrently as a Blockchain Developer at Okamalabs (from mid-2021) and under the amakodev personal brand — applying coursework directly to real-world blockchain and software development throughout the degree programme.

CERTIFICATIONS & LEARNING

Alison Diplomas & Professional Development

- ◆ Diploma in Software Project Management
- ◆ Diploma in Cyber Security
- ◆ Diploma in Machine Learning with Python
- ◆ Advanced Diploma in Modern Application Development
- ◆ Diploma in Python Programming
- ◆ Diploma in Applied Generative AI

Focus: Continuous self-directed learning across software project delivery, modern application development, cybersecurity, Python engineering, machine learning, and applied generative AI.

PROFESSIONAL PROFILES

LINKEDIN [linkedin.com/in/amakodevz](https://www.linkedin.com/in/amakodevz)

GITHUB github.com/amakodev

PORTFOLIO amsoftwarezw.space

UPWORK [upwork.com/freelancers/~012553b7bf3e196c60](https://www.upwork.com/freelancers/~012553b7bf3e196c60)

ADDITIONAL INFORMATION

WORK ETHIC

Known for taking full ownership of projects, working with clarity, and delivering complete solutions rather than partial implementations.

COLLABORATION STYLE

Comfortable working with founders, startups, and product teams to turn early-stage ideas into practical software systems with clear execution and product-minded delivery.

LEARNING MINDSET

Continuously develops practical skills in AI automation, SaaS architecture, and modern full-stack product delivery through self-directed learning and applied project work.

INTERESTS

Maintains an active lifestyle through gym training, football, martial arts, hiking, and reading — habits that reinforce discipline, consistency, resilience, and focus.